Source-Based AI Recommendation Explanations — Hypotheses v1

Research Question

How does the information source disclosed in an AI-generated recommendation explanation influence consumers' perceptions of relevance, credibility, privacy, trust, and willingness to rely on the recommendation?

H1 — Perceived Relevance

- **H1a.** A behavior-based recommendation explanation will lead to higher perceived relevance than a no-source control explanation.
- **H1b.** A product-feature-based recommendation explanation will lead to higher perceived relevance than a no-source control explanation.

Rationale

Behavior-based explanations explicitly connect the recommendation to the user's prior behavior, while product-feature-based explanations explicitly connect product attributes to the user's stated preferences. Both therefore provide a personalized rationale for why the recommended product fits the individual user.

H2 — Privacy Concern

- **H2.** A behavior-based recommendation explanation will produce greater privacy concern than review-based, product-feature-based, and no-source explanations.

Rationale

Behavior-based explanations explicitly reveal that the system has used the consumer's prior browsing or shopping activity. This personalization may increase perceived relevance, but it can simultaneously make the collection and use of personal behavioral data more salient.

H3 — Source Credibility

- **H3.** A review-based recommendation explanation will produce greater perceived source credibility than a no-source explanation.

Rationale

A review-based explanation makes an identifiable external information category—other consumers' evaluations—salient as evidence supporting the recommendation. Consumer reviews can therefore provide an additional informational basis for evaluating the recommendation rather than requiring the user to rely solely on the AI system's assertion.

H4 — Psychological Mechanisms of Trust

- **H4a.** Perceived relevance will be positively associated with trust in the AI recommendation.
- **H4b.** Perceived source credibility will be positively associated with trust in the AI recommendation.
- **H4c.** Privacy concern will be negatively associated with trust in the AI recommendation.

H5 — Trust and Reliance

- **H5.** Greater trust in the AI recommendation will be associated with greater willingness to rely on the recommendation.

H6 — Indirect Effects of Explanation Source

- **H6.** The effect of recommendation explanation source on reliance will operate indirectly through perceived relevance, source credibility, privacy concern, and subsequent trust.

Exploratory Question — Mixed-Source Explanation

- **RQ1.** How does a mixed-source explanation combining behavioral information, customer reviews, and product-feature information affect perceived relevance, source credibility, privacy concern, trust, and reliance relative to single-source explanations?

No directional hypothesis is specified for the mixed-source condition in the initial study.